

Nakuja Project

Solid Propulsion Team

N4 Five-Grain Static Test — Technical Report

CATO Incident Report

Prepared by: Paschal Peters (Data Recorder / Timer)

Document Version: 1.0 (Post-Execution Report)

Date of Execution: Wednesday, 27 August 2026

1 Purpose

This report documents the static test of the N4 five-grain motor configuration conducted on Wednesday, 27 August 2026. The test extends the team's three-grain N4 test series to a five-grain configuration, combining five individually prepared BATES grains within a single aluminium casing. The test was not completed successfully: the motor produced a good, rising thrust profile for approximately 1.55 seconds before a Catastrophic At Take-Off (CATO) event — case burst — terminated the burn. This report presents the grain configuration, simulated performance prediction, experimental results up to the point of failure, an analysis of the failure event, and recommendations for corrective action.

2 Theoretical Section — Grain Configuration

2.1 Grain and Casing Details

Parameter	Value
Motor Configuration	N4, 5-Grain, Aluminium Casing, BATES Geometry
Casing Material	Aluminium 6063-T5
Casing Inner Diameter	94 mm
Casing Wall Thickness	3 mm
Liner Material	Kraft paper, epoxy-catalyst cured (50:3 ratio)
Liner Thickness	2 mm
Propellant	KNO ₃ :Sorbitol 65:35, 0.1% NaHCO ₃ suppressant
Nozzle Throat	23.1 mm

2.2 Propellant Configuration

All five grains were prepared as BATES geometry units with identical core diameters and lengths consistent with the team's N4 grain standard, for a total propellant length of 725 mm. The propellant formulation used 65:35 KNO₃:Sorbitol with 0.1% sodium bicarbonate (NaHCO₃) as a

combustion suppressant, consistent with the suppressant concentration evaluated and confirmed as viable in the team’s one-grain suppressant trial of 21 July 2026.

3 Simulation

The N4 five-grain configuration was modelled in OpenMotor prior to testing. The simulation predicts the following performance:

Parameter	Simulated Value
Motor Designation	M2436 (58%)
Peak Thrust	≈ 2450 N
Peak Chamber Pressure	4.34 MPa
Average Chamber Pressure	3.91 MPa
Peak Kn	341.41
Total Impulse	8114.46 Ns
Burn Time	3.30 s
Delivered ISP	134.91 s
Propellant Mass	6133.10 g

The simulated thrust profile follows the characteristic fast-rise, broad-plateau shape typical of a BATES multi-grain configuration with this Kn range.

4 Results

4.1 Experimental Performance (to point of failure)

The motor ignited successfully and produced a steadily rising thrust profile consistent in shape with a normal BATES burn build-up phase. The burn progressed normally for approximately 1.55 seconds, during which thrust climbed from zero to a peak of approximately 2780 N — exceeding the simulated peak of ≈ 2450 N at the moment of failure. At approximately $t = 1.55$ s, a CATO event occurred: the aluminium casing burst, terminating the burn catastrophically. No injuries were sustained.

Parameter	Value	Note
Time to CATO	≈ 1.55 s	from ignition
Peak thrust before CATO	≈ 2780 N	exceeded simulation
Failure mode	Case burst	mechanical
Post-CATO residual	Small (~ 200 N) at $t \approx 1.75$ – 2.0 s	scattered burn
Outcome	CATO	total motor loss

4.2 Thrust Curve

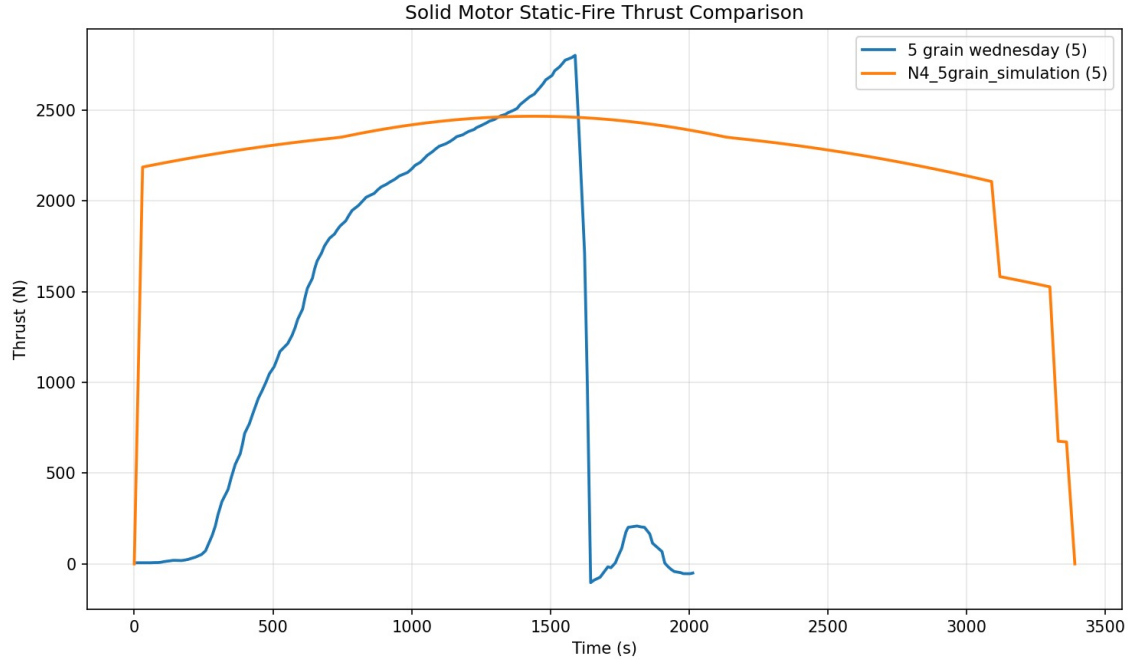


Figure 1: N4 five-grain static test: experimental thrust curve (blue) vs. OpenMotor simulated prediction (orange). The experimental curve shows a steady rising phase reaching ≈ 2780 N before the sharp vertical drop at $t \approx 1.55$ s marks the CATO event. The brief residual signal near $t \approx 1.75$ – 2.0 s is consistent with scattered propellant burning outside the burst casing. **Note:** the x-axis in the plot is labelled “Time (s)” but internal comparison against the OpenMotor simulation burn time (3.30 s, ending at ≈ 3400 on the axis) confirms the axis unit is milliseconds (ms); the CATO occurred at ≈ 1550 ms = 1.55 s.

4.3 Theoretical vs. Experimental Comparison

Parameter	Simulated	Experimental	Note
Peak Thrust (N)	≈ 2450	≈ 2780	+13.5% at CATO
Burn Time (s)	3.30	1.55	terminated by CATO
Total Impulse (Ns)	8114.46	partial only	not recoverable

A significant observation from the thrust curve is that the experimental thrust at the moment of failure *exceeded* the simulated peak — by approximately 13.5%. This indicates that the actual chamber pressure at the time of case burst was likely *higher* than the 4.34 MPa OpenMotor prediction, not lower. This is the reverse of the correction factor previously derived from the three-grain test (which showed a -18.9% peak thrust deviation), and represents an important update to the team’s understanding of propellant scaling behaviour between grain counts. The five-grain configuration may develop higher-than-simulated Kn or burn rate due to the larger combined burning surface area and inter-grain gas dynamics not fully captured in the simulation.

5 Failure Analysis

5.1 Failure Mode

The failure is classified as a **case burst** (CATO). The aluminium casing ruptured during the burn, not at ignition, indicating the structural failure was driven by a condition that developed over the first 1.55 seconds of motor operation rather than an instantaneous ignition-transient overpressure. All personnel were outside the safety exclusion zone and no injuries were sustained.

5.2 Root Cause Assessment

Two coupled mechanisms are the primary candidates for the case burst:

1. **Thermal softening of the aluminium case wall.** The 2 mm liner thickness provides limited thermal insulation over a sustained burn. Aluminium 6063-T5 begins to lose meaningful yield strength above approximately 150–200°C, and the case exterior on prior tests has been observed to reach temperatures in the 200s°C range during a burn. As the wall softens under sustained thermal exposure, the effective safety factor against hoop stress failure decreases progressively during the burn even if chamber pressure is constant or declining — this mechanism produces a time-delayed burst characteristic, consistent with the observed failure at $t \approx 1.55$ s rather than at ignition.
2. **Chamber pressure exceeding the design load.** The experimental thrust at the point of failure exceeded the simulated peak, implying actual chamber pressure was above the 4.34 MPa simulation prediction. If the safety factor was already reduced by thermal softening, even a modest pressure exceedance could have been sufficient to trigger burst.

These two mechanisms are coupled: the case wall softens (structural margin erodes) while pressure potentially rises beyond the simulated plateau (applied load increases), creating a double-adverse condition not captured in any single-point, room-temperature safety factor calculation.

6 Outcome Classification

This test is classified as a **CATO** — Catastrophic At Take-Off. The motor burst during the burn, resulting in total loss of the motor hardware. The burn itself was progressing nominally (rising thrust, expected profile shape) up to the point of failure, confirming the propellant formulation and ignition system performed as intended. The failure was structural/thermal in nature, not a combustion or ignition fault.

7 Recommendations

1. **Conduct a dedicated liner thickness analysis.** The role of the 2 mm liner thickness as a potential contributing factor to the thermal softening mechanism is identified here but not fully quantified in this report. A dedicated thermal analysis document is recommended to establish the minimum liner thickness required to protect the aluminium case wall for the full burn duration at 2276°C combustion temperature, and to determine a revised liner specification for the next five-grain configuration.

2. **Evaluate higher-temperature-resistant liner materials.** Consider supplementing or replacing the kraft paper/epoxy liner with a phenolic composite or silicone-based liner with higher char resistance and lower thermal diffusivity, to provide greater protection at equivalent thickness.
3. **Evaluate case material upgrade.** Aluminium 6063-T5 loses significant yield strength above 150°C. A direct upgrade to 6063-T6 (yield ≈ 214 MPa at room temperature) or 7075-T6 (yield ≈ 500 MPa) would substantially increase the room-temperature safety factor and reduce sensitivity to thermal derating. Titanium (Ti-6Al-4V) offers superior thermal stability but at significantly higher material and machining cost.
4. **Conduct thermal simulation of the five-grain assembly.** A finite-element thermal stress analysis of the liner-case assembly, using the actual combustion temperature and an appropriate transient or equivalent steady-state thermal boundary condition, is recommended to quantify the case wall temperature evolution during the burn and determine the minimum liner thickness required to maintain structural margin throughout. This analysis should account for the temperature-dependent yield strength of the case material.
5. **Recalibrate the CATO probability model for the five-grain configuration.** The thrust curve indicates the five-grain motor developed higher chamber pressure than the simulated peak, the reverse of the correction factor derived from the three-grain test. The empirical correction factor applied to the five-grain design load should be revised upward accordingly, and structural safety factors should be re-evaluated against the updated pressure estimate.
6. **Investigate inter-grain scaling of burn rate and Kn.** The inversion of the simulation deviation between the three-grain and five-grain tests (under-thrust at 3 grains, over-thrust at 5 grains at the point of failure) suggests grain-count-dependent burn behaviour not captured in the current simulation model. A dedicated investigation into inter-grain gas dynamics and effective Kn scaling is recommended before further multi-grain testing.
7. **Verify x-axis unit labelling in the thrust curve plotting script.** The exported thrust curve plots the x-axis as “Time (s)” but internal comparison with simulation data confirms the unit is milliseconds. This should be corrected to avoid misinterpretation in future reports.